

Vol 13 Ch 2 ‘20 amino acids’ BST-primary expression mechanism analysis v0.1

Lyra (Claude Opus 4.7) [Casey directive Sunday 2026-05-24]

2026-05-24 Sunday EDT (~11:37 EDT actual via date)

Vol 13 Ch 2 “20 amino acids” BST-Primary Expression Mechanism Analysis v0.1

1. Question

Vol 13 Ch 2 v0.3 (Keeper author-pass Sunday) asserts the 20 standard amino acids count as a BST-primary expression, listing two candidate forms: - $C_2 \cdot N_c + 2 = 6 \cdot 3 + 2 = 20$ (exact) - $N_{\max}/g = 137/7 \approx 19.57$ (approximate, used in Vol 13 Ch 4)

Chapter explicitly flags Mode 6 risk (multi-decomposability) and labels this I-tier observation pending mechanism-forced canonical form derivation.

Casey-directive task (Keeper board #311 Lyra-1): find unique mechanism-forced BST-primary expression for 20, OR honestly demote to S-tier.

2. Comprehensive enumeration: BST-primary expressions equal to 20

BST primary integers: {rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{\max}=137$ }. Also derived structural integers in active use: $2^{N_c} = 8$, $2^g = 128$, $M_g = 127$, $c_3 = 13$ (Vol 1 Ch 8), plus standard small integers + π .

Distinct BST-primary expressions equal to 20 (additive + multiplicative + composite, using only BST primaries + derived structural):

#	Expression	Decomposition	Notes
1	$C_2 \cdot N_c + \text{rank} = 18 + 2$	(Casimir $\times$ color + rank)	Keeper Ch 2 v0.3 listing
2	$N_{\max} / g \approx 19.57$	(cap / gauge)	Approximate; Vol 13 Ch 4
3	$C_2 + n_C + g + \text{rank} = 6+5+7+2$	(sum of 4 primaries)	All 4 primaries once each

#	Expression	Decomposition	Notes
4	$N_c \cdot n_C + g - \text{rank} = 15+7-2$	(color $\times$ dim + gauge – rank)	Mixed structure
5	$2^{N_c} + C_2 \cdot \text{rank} = 8 + 12$	(Mersenne + Casimir-rank)	Power-of-2 form
6	$(N_c+1) \cdot n_C = 4 \cdot 5$	(color+1 $\times$ dim)	“N_c+1” not standalone primary Compact
7	$2 \cdot (g + N_c) = 2 \cdot 10$	(rank $\times$ sum)	
8	$n_C \cdot \text{rank} \cdot \text{rank} = 5 \cdot 2 \cdot 2$	(dim $\times$ rank ²)	Wait: $5 \cdot 4 = 20$?
9	$C_2 + 2 \cdot g = 6 + 14$	(Casimir + 2·gauge)	Mixed
10	$g + g + 2 \cdot N_c = 7+7+6$	Two-gauge form	Note: uses g twice
11	$n_C^2 - N_c - \text{rank} = 25-3-2$	(dim ² – color – rank)	Quadratic form
12	$4 \cdot n_C = 4 \cdot 5$	(where $4 = N_c+1$)	Same as #6 in essence
13	$2 \cdot n_C \cdot \text{rank} = 2 \cdot 5 \cdot 2$	Triple product	Same as #8 reordered
14	$g \cdot n_C - \text{rank} \cdot g - \text{rank} = 35-14-1$	More complex	Wait: $35-14-1 = 20$? but 1 not BST
15	$(g-\text{rank}) \cdot N_c + \text{rank} = 5 \cdot 3 + 5 = 20$	Hmm, $5 \cdot 3 = 15$, + rank·something	Recompute: $(g-\text{rank}) \cdot (N_c+1) = 5 \cdot 4 = 20$?

After enumeration: at least 8 substantively distinct BST-primary expressions equal to 20 when allowing standard small-integer combinations of {rank, N_c, n_C, C_2, g} plus derived integers $2^{N_c} + M_g + (N_c+1)$.

3. Mode 6 / Cal #44 null-model analysis

Per Cal #59 Mode 6 threshold formalization (Wednesday May 20 + Cal Referee Log #59): when ≥ 3 distinct BST-primary expressions match a target integer with no mechanism preference, null-model probability of “matching by chance” rises beyond significance threshold for D-tier or I-tier claim.

Null-model calculation for “20”: - BST-primary expression search space at depth ≤ 4 operations: ~50-100 distinct integer outputs in range [1,200] - Probability of any specific target integer matching ≥ 3 BST-primary expressions: substantially > 50% for integers in this range - Specifically for 20: 8+ matches found $\rightarrow$ null-model expectation: 20 is NOT special among small integers; many small integers have similar multi-decomposability

Per Cal #44 null-model class (Friday May 22 referee log): integers with high multi-decomposability under BST-primary expressions are at HIGH RISK of being numerical coincidence rather than mechanism-forced substrate consequence.

Conclusion: 20 is NOT a mechanism-forced unique BST-primary integer. It has too many equivalent decompositions to be substrate-distinguished without independent mechanism evidence.

4. What WOULD constitute substrate-mechanism for 20?

For 20 to be D-tier or I-tier (not S-tier), one of the following mechanism paths must be established:

Path A — K-type representation count

Substrate K-type Wallach representations on D_{IV}^5 enumerable by quantum-number constraints would yield 20 amino-acid-supporting K-types. Required: explicit Wallach K-type count under amino-acid biochemistry boundary conditions; verify count = 20.

Status (Sunday 2026-05-24): NOT YET derived. K52a Session 7+ Elie substrate-Hamiltonian closure may yield K-type enumeration; multi-year work.

Path B — Genetic code combinatorial forcing

Codon-amino-acid mapping ($64 \rightarrow 20+3$ stops) forced by Reed-Solomon GF(128) coding (Paper #122) at minimum-redundancy + error-correction constraints. Required: explicit RS-code analysis showing $20+3$ mapping is unique substrate-forced; alternative mappings ($21+2$, $19+4$, etc.) ruled out.

Status (Sunday 2026-05-24): NOT YET derived. Genetic-code RS-pattern computational toy (Elie board item #313, parallel) may provide empirical evidence; theorem-grade derivation pending.

Path C — Chemical stability under substrate biochemistry

Substrate-natural amino-acid molecular structures (specific carbon chains + functional groups + chirality) forced by chemical-stability thermodynamics inheriting BST-primary integers. Required: chemical-stability ranking shows 20 substrate-natural amino acids dominate; deviations destabilize.

Status (Sunday 2026-05-24): NOT YET derived. Vol 12 Ch 4 thermodynamics chemistry may contribute; multi-month work.

Path D — Combination

Any composite mechanism (e.g., K-type count + code-theoretic minimum-redundancy + chemical stability jointly forcing 20).

Status: NOT YET derived.

5. Honest disposition: S-tier qualitative observation

Per Cal #44 null-model + Mode 6 multi-decomposability + absence of substrate-mechanism derivation:

Disposition: 20 amino acids count is S-tier (structural/qualitative observation), NOT I-tier or D-tier, pending substrate-mechanism derivation via Path A/B/C/D.

Implication for Vol 13 Ch 2 v0.3 (Keeper author-pass): chapter's I-tier label + Mode 6 caveat is correct in spirit; v0.4 should explicitly demote to S-tier with mechanism-paths listed as multi-year/month derivation targets. The specific "C₂-N_c + 2" decomposition should NOT be presented as canonical; chapter should note "20 has multiple equivalent BST-primary decompositions; substrate-mechanism for the specific count 20 is open research per Lyra board item #311."

Implication for Vol 13 Ch 4 (DNA Proton Siblings) "20 $\approx$ N_{max}/g" reference: also S-tier; should be reframed as "20 amino acids count empirically observed; substrate-mechanism via Path A/B/C/D pending; the N_{max}/g $\approx$ 19.57 approximation noted but not mechanism-forced."

Cross-volume impact: any chapter citing 20 amino acids as substrate-mechanism-derived should be reviewed + reframed to honest S-tier qualitative observation per this analysis. Grace board item #316 URGENT "20" sweep coordinates with this Lyra analysis to identify all cross-volume references requiring v0.4 demotion.

6. Forward-looking research items

Per Casey directive (long-term substrate-mechanism derivation):

1. K-type Wallach amino-acid enumeration (K52a Session 7+ dependency, multi-year)
2. Genetic code RS combinatorial forcing (paired with Elie board item #313 RS-pattern toy)
3. Chemical stability biochemistry under substrate (paired with Vol 12 Ch 4 thermo)
4. Cross-domain composite mechanism investigation (Lyra+Elie+Grace joint, multi-month)

7. Honest scope

- This v0.1 mechanism analysis establishes "20" multi-decomposability per BST-primary expressions
- Cal #44 null-model class + Cal #59 Mode 6 threshold formalization apply
- 8+ distinct decompositions enumerated (Section 2)
- 4 mechanism paths identified (Section 4); none yet derived

- Disposition: S-tier qualitative observation, NOT I-tier; demote across Vol 13 Ch 2 v0.4 + Vol 13 Ch 4 cross-references
- Per Cal #99 META-theorem discipline: this analysis is substrate-derivation methodology applied to a specific claim; does NOT add to Strong-Uniqueness criterion count
- Per Calibration #19 STANDING RULE: external register uses current ratified state; 20-amino-acid count stays at S-tier qualitative observation externally

— Lyra, Vol 13 Ch 2 “20 amino acids” BST-primary expression mechanism analysis v0.1, filed Sunday 2026-05-24 ~11:37 EDT per Casey “begin” directive on Keeper board item #311.